Examiner only

11. Seawater contains a number of dissolved salts. Although composition varies with location, 1 000 cm^3 of seawater contains about 20 g of chloride ions, Cl^-, and about 3 g of sulfate ions, SO_4^{2-}.

A student is given a sample of seawater from Rhossili Bay and asked to determine the chloride ion content by volumetric analysis and the sulfate ion content by gravimetric analysis.

(a) Determination of chloride ion content by volumetric analysis.

The method is similar to an acid-base titration. A silver nitrate solution of known concentration is used to precipitate chloride ions as silver chloride.

$$Ag^+(aq) \ + \ Cl^-(aq) \longrightarrow AgCl(s)$$

The seawater is diluted by a factor of five before it is used in the titration.

The endpoint of this titration is difficult to determine directly, so potassium chromate(VI), K_2CrO_4, is used as an indicator. When all of the chloride ions have been used up, the chromate(VI) ions react with silver ions and produce silver chromate(VI), which forms a red precipitate. The instant a permanent red tinge appears in the solution, the endpoint has been reached.

Volume of diluted seawater in the conical flask = 25.0 cm^3

Concentration of silver nitrate solution in the burette = 0.100 mol dm^{-3}

Mean titre = 26.40 cm^3

(i) Before starting the titration, the student rinses the burette with silver nitrate solution. Suggest why he does this. [1]

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(ii) Suggest why the student dilutes the seawater. [1]

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(iii) Describe how the student should dilute the seawater by a factor of five. [3]

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(iv) Describe and explain **one** action the student might take just before the endpoint of the titration, to ensure that the volume of silver nitrate added at the endpoint is accurate. [2]

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(v) Write an ionic equation for the precipitation of silver chromate(VI). [1]

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(vi) Calculate the mass of chloride ions in 1000 cm^3 of the original seawater, giving your answer to an **appropriate** number of significant figures. [4]

Mass of chloride ions = ... g

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(b) Determination of sulfate ion content by gravimetric analysis.

> $100\,cm^3$ of undiluted seawater and $0.100\,mol\,dm^{-3}$ barium nitrate solution were used.
>
> The mass of the barium sulfate precipitate was 0.65 g.
>
> You may assume that **all** of the sulfate ions in the seawater were precipitated.

(i) Describe how the student carried out the gravimetric analysis to find the mass of the barium sulfate precipitated. [5]

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(ii) Calculate the minimum volume, in cm^3, of barium nitrate solution needed to precipitate all of the sulfate ions in $100\,cm^3$ of the seawater. [3]

Volume = ... cm^3

(iii) Suggest why the volume of barium nitrate needed was different to the volume of seawater used. [1]

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END OF PAPER